

I2C	Sub SYS	Function	Part Number	I2C Spec.	Budget Timing	I2C Slave Address / Write / Read (7-bit mode)	
I2C-0	AP	Cap Touch controller	GT1151	400 Kbps	Yes.	0x50	Write 0x6A / Read 0x6B
		LCM Gate Drive	RT4801HWSC	400 Kbps	Yes.		
I2C-1 (BC)	AP Sensor Hub	Magnetic Sensor	AK09918C	400 Kbps	Yes.	0x0C	Write 0x18 / Read 0x19
		Ambient Light Sensor	CM36558	400 Kbps	Yes.	0x51	Write 0xA2 / Read 0xA3
		Proximity Sensor					
		Pressure Sensor	BMP280	400 Kbps	Yes.	0x77	Write 0xE6 / Read 0xEF
I2C-2 (BC)	AP	Rear Camera	IMX519	400 Kbps		0x1A	Write 0x34 / Read 0x35
			EEPROM	400 Kbps		0x20	Write 0xA2 / Read 0xA1
		AF driver		400 Kbps		0x0C	Write 0x18 / Read 0x19
		Dual cam (B+B) - Main1	ISIC091	400 Kbps		0x20	Write 0x34 / Read 0x35
			EEPROM	400 Kbps		0x50	Write 0xA2 / Read 0xA1
		AF driver		400 Kbps		0x0C	Write 0x18 / Read 0x19
I2C-3	AP	NFC	ST21NFCD	400 kbps	Yes.	0x08	Write 0x10 / Read 0x11
		Dual cam (B+B) - Main2	OV8856	400 Kbps		0x10	Write 0x20 / Read 0x21
			EEPROM/HNA				
		AF driver/HNA					
I2C-4 (BC)	AP	Front Camera	ISIC217	400 Kbps		0x20	Write 0xA / Read 0x0B
			EEPROM	400 Kbps		0x52	Write 0xA4 / Read 0xA5
I2C-5	AP	Reserved					
I2C-6	AP SSPM	Reserved					
I2C-7	AP	Audio Smart PA	MT6980	400 Kbps	Yes.	0x34	Write 0x68 / Read 0x69
		SW Charger	SW Charger	400 Kbps	Yes.		
		CC Detect	RT11715WSC	400 Kbps	Yes.		
		Flash LED SV Boost	RT4505	400 Kbps	Yes.		

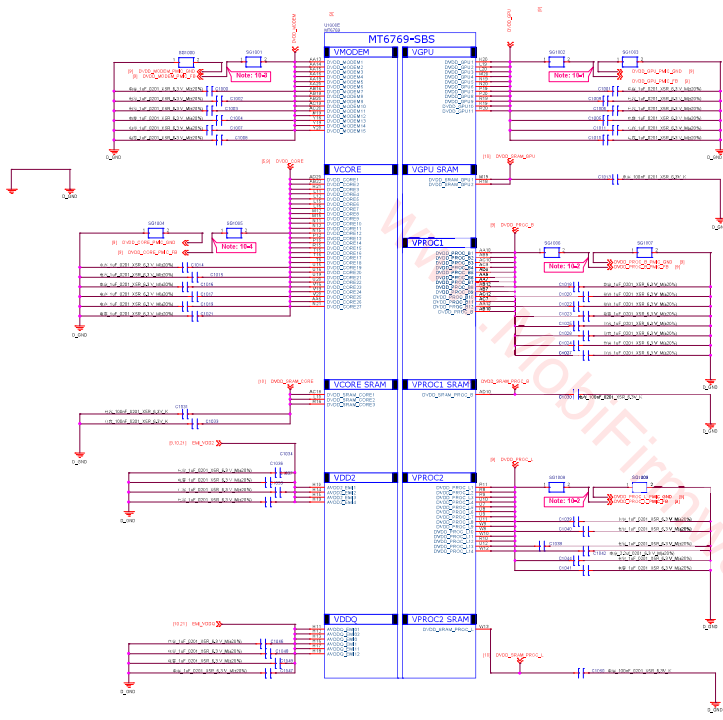
Note : I2C Spec. : Standard mode (100 kbps) and Fast mode (400 kbps), Fast mode Plus (1 Mbps) and High-speed mode (3.4 Mbps)

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Date	Category	Item
2018.12.14 (V0.1)	Page 04	V0.1 Release
2019.01.11 (V0.2)	Page 12	Add CAM_PDN3(GPIO110) HW pin For Note 12-4: HW pin for DDR type feature in bootstrap Delete R1007 and R1012 ext. pull for LP4X eMCP
	Page 13	Add CAM_PDN3(GPIO110)'s Note : No ext. pull for DDR type feature in bootstrap
	Page 69	Add CAM_PDN3(GPIO110)'s Note : No ext. pull for DDR type feature in bootstrap
	Page 21	Add C2106 for VBIF28 cap
	Page 22	Delete R2202

www.MobiFirmware.com

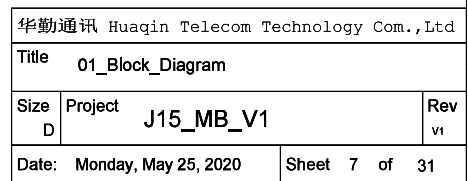
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Schematic design notice of "10_BB_POWER_PDN" page:

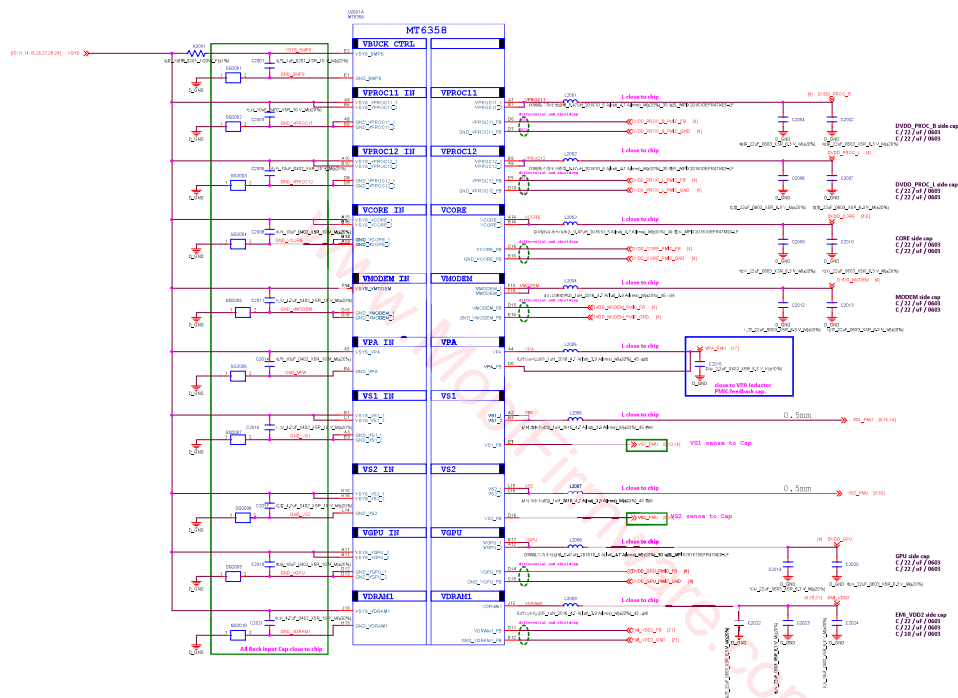
- Note 10-1: Differential pair of DVDD_GPU remote sense must be close to BB's tail.
Remote sense trace with GND shielding to PMIC (Differential)
- Note 10-2: Differential pair of DVDD_PROC remote sense must be close to BB's tail.
Remote sense trace with GND shielding to PMIC (Differential)
- Note 10-3: Differential pair of DVDD_MODEM remote sense must be close to BB's tail.
Remote sense trace with GND shielding to PMIC (Differential)
- Note 10-4: Differential pair of DVDD_CORE remote sense must be close to BB's tail.
Remote sense trace with GND shielding to PMIC (Differential)

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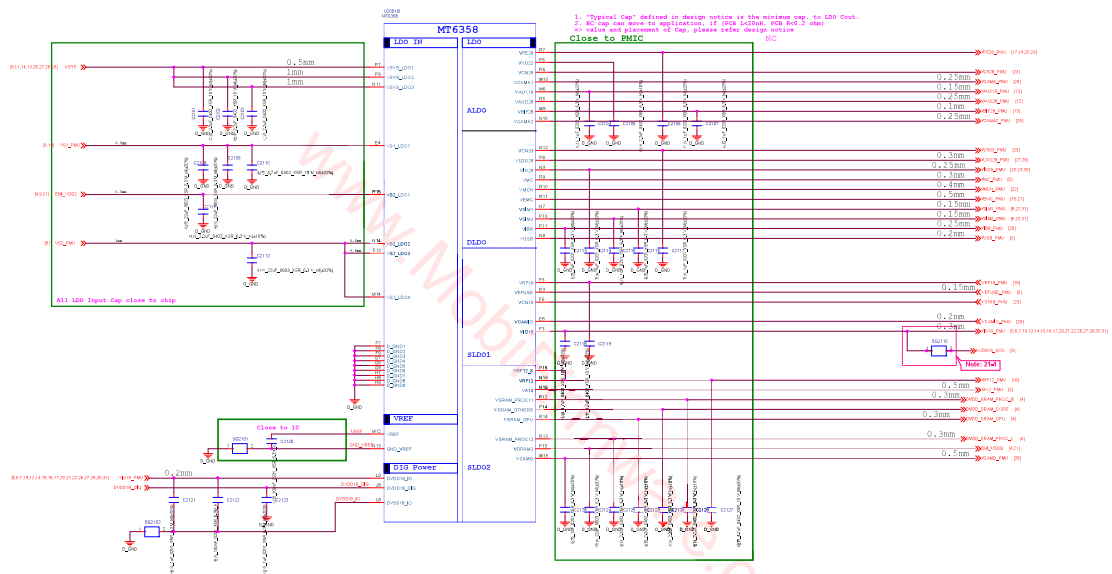




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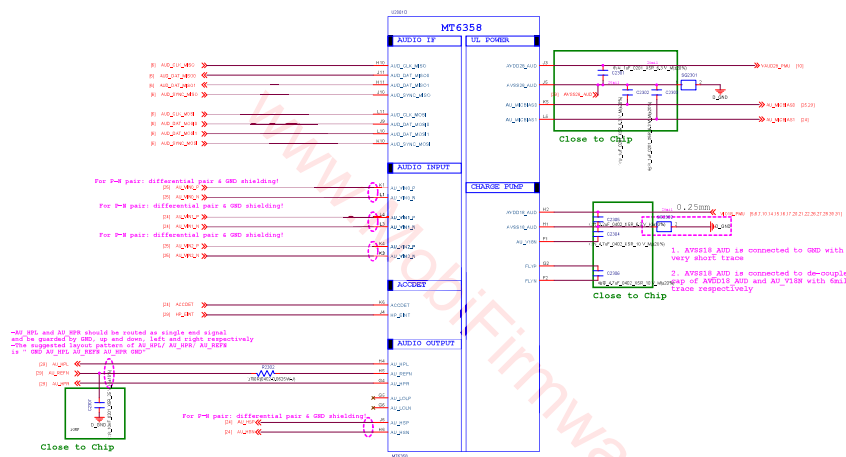


Schematic design notice of "21_POWER_MT6358-LDO" page.

Note 21-1: Please set 502110 close to C2118, making star connection between VQ18_PMU and AVDD18_SOC near to LDO cap, C2118. Please also refer to MT6358 design notice for further detail design information.

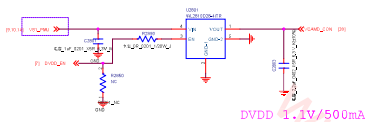
Note 21-2: If these power trace can meet LDO layout constraint, these CAP can be NC or removed. Please refer to MT6358 design notice.

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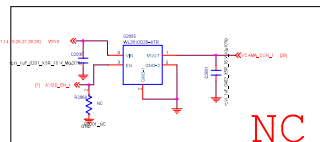


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Title O1_Block_Diagram			
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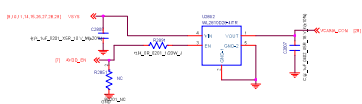
LDO for Rear Camera (VCAMD_EXT1)
1.06V



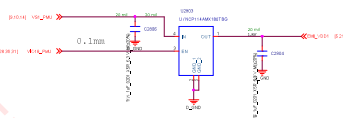
**LDO for Dual Camera (VCAMA_EXT)
2.8V**



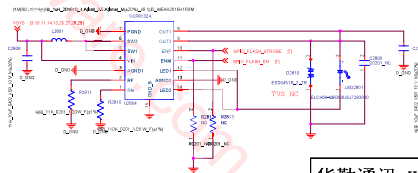
**LDO for Dual Camera (VCAMA_EXT)
2.8V**



Ext. EMI_VDD1 for LPDDR4X VDD1



Flash LED 5V Boost



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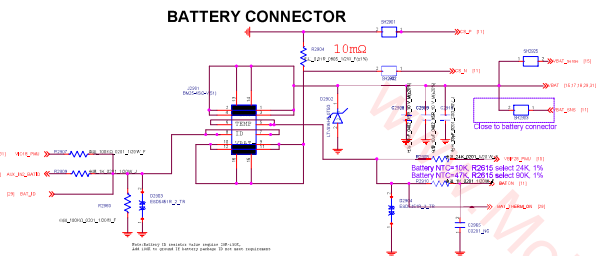
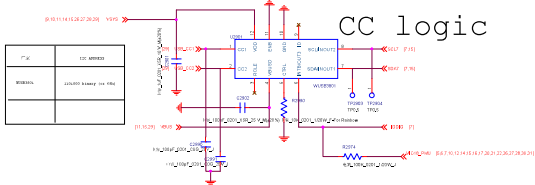
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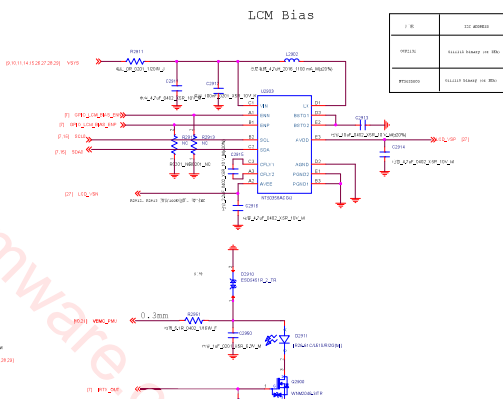
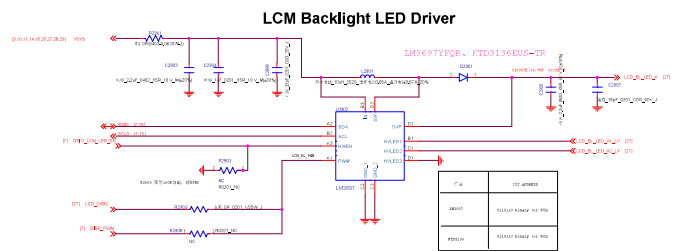
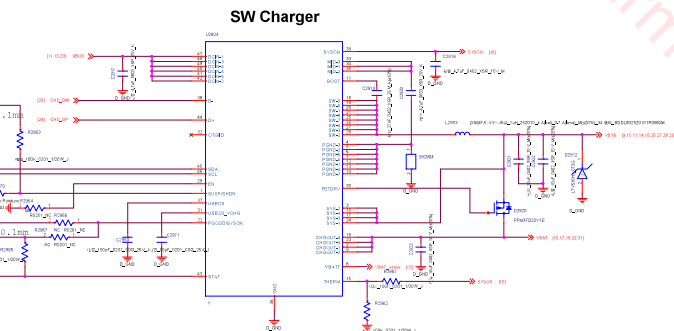
Rev
v1

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Schematic design notice of "28_EXT_WLED+DB+FL+CHG+PD" page.
 Note 28-1: For better ESD or surge performance we need choose suitable device for system protection, Please refer to [Surge device selection guide V2.0] provide by MTK.



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Transceiver

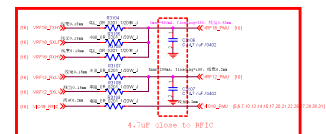
Attenuator for Detector

The lower detection gain is 0.70dB, coupling factor is 0.01dB.

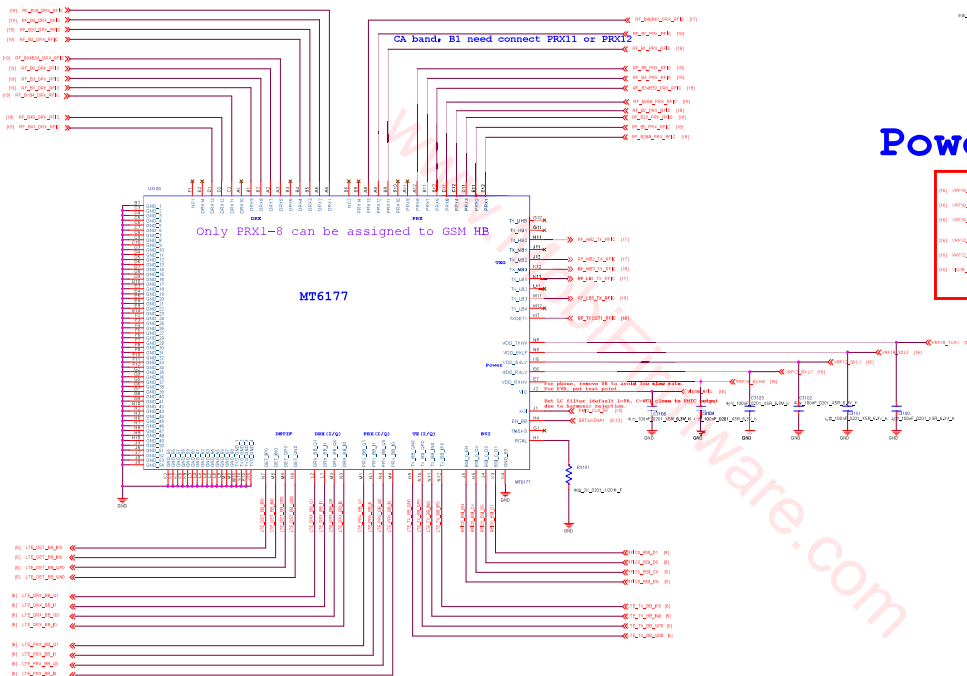
4dB Attenuator



Power For MT6177



GND



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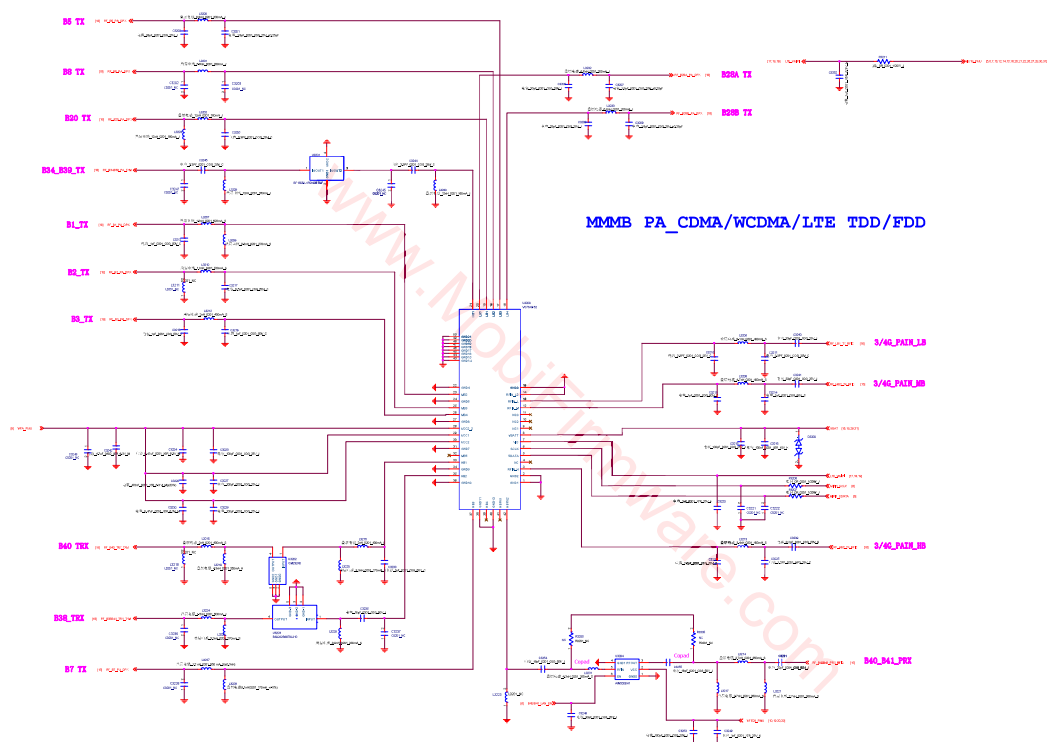
Title 31_RF_MT6177_Pin_Out

Size D Project F9_MB_V1

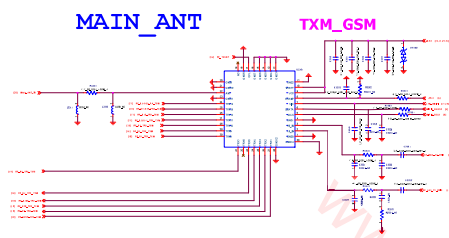
Rev v1

Date: Monday, May 25, 2020

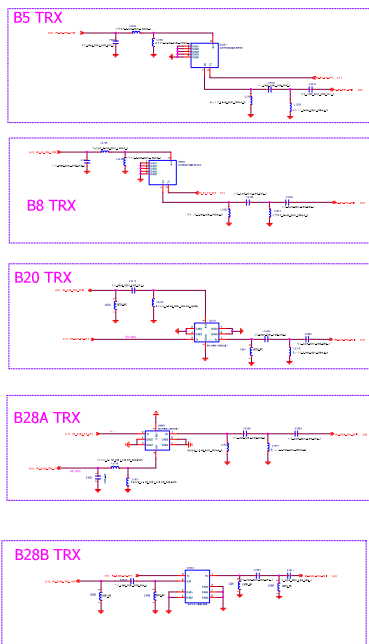
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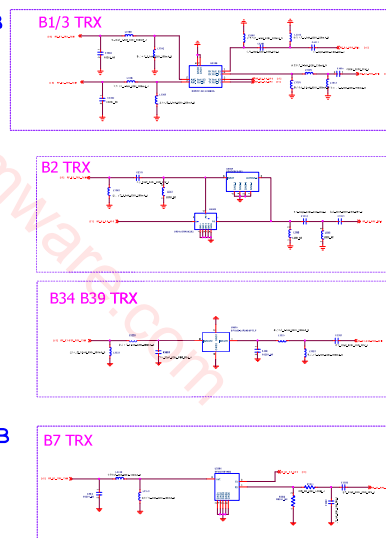
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 32_RF_MT6177_RF_TX			
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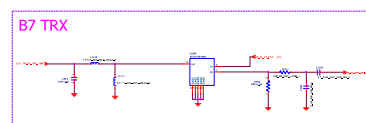
TRX_LB



TRX_MB

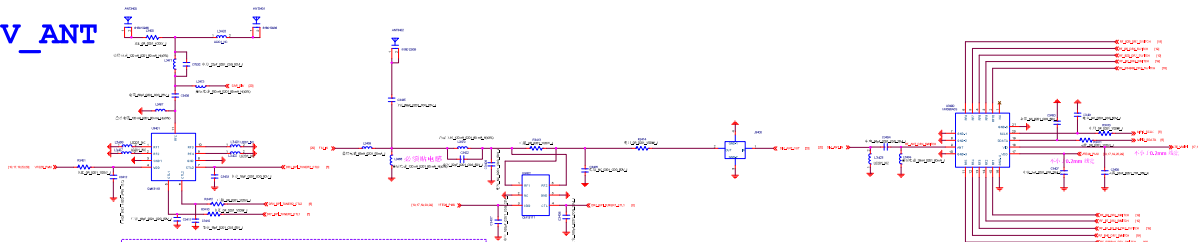


TRX_HB

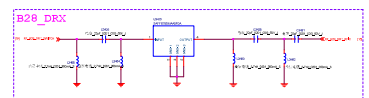
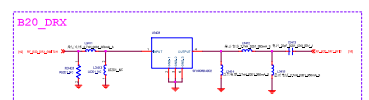
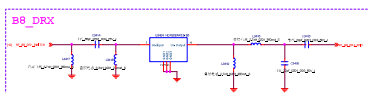
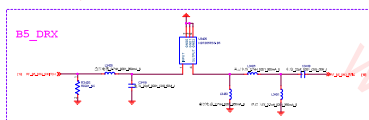


华勤通信 Huqin Telecom Technology Co., Ltd			
Title 33_RF_MT8177_RF_PRR			
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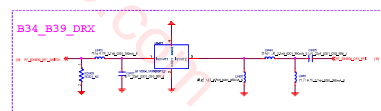
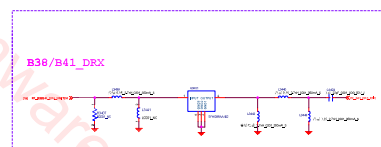
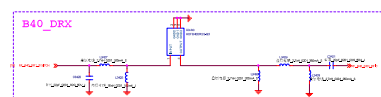
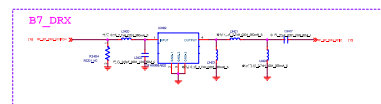
DIV_ANT



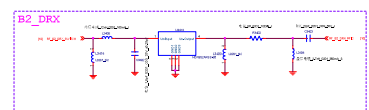
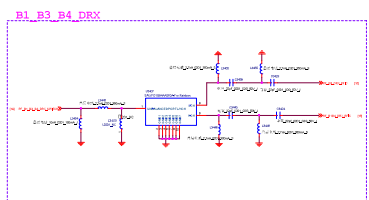
DIV_LB



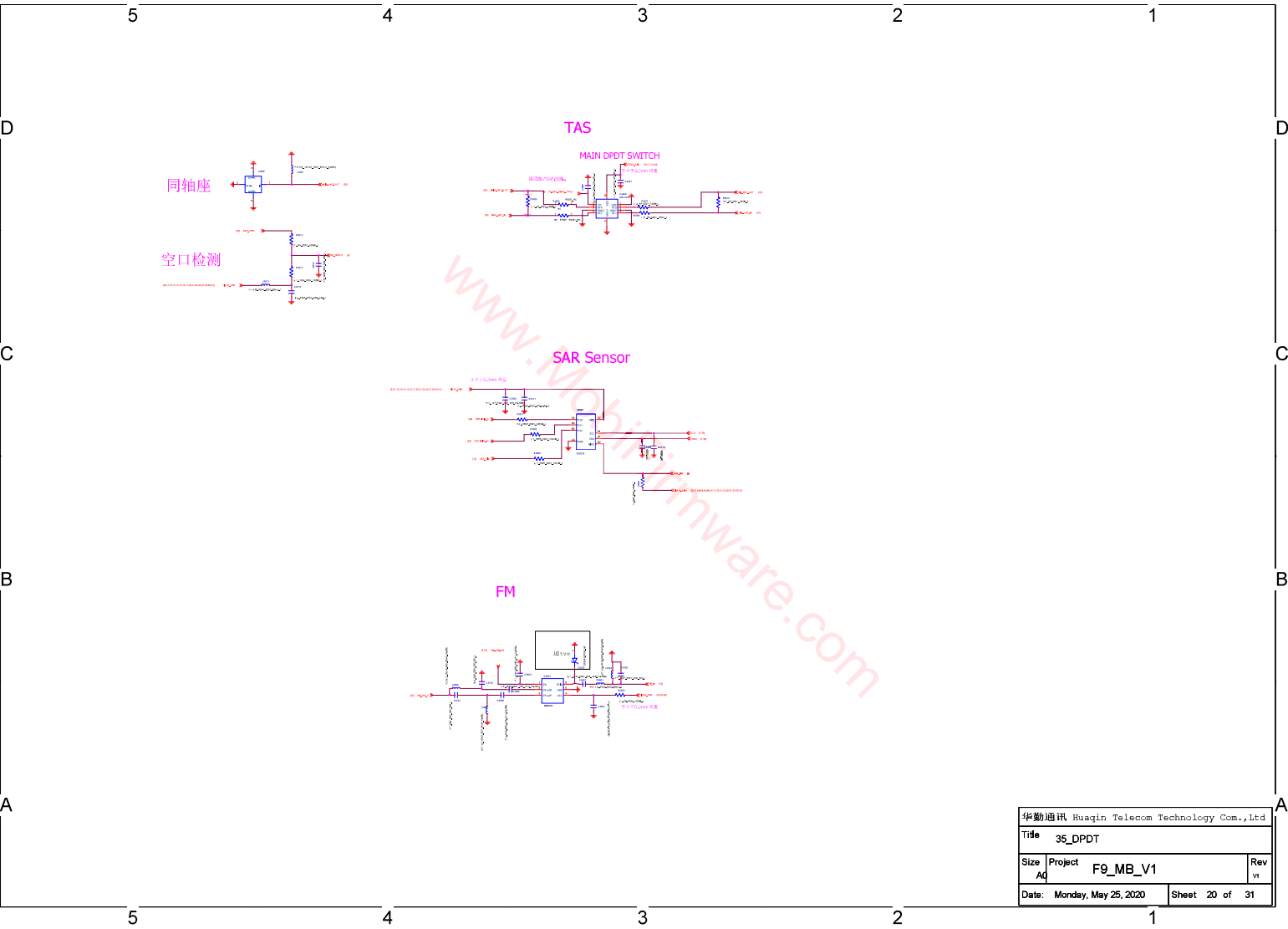
DIV_HB

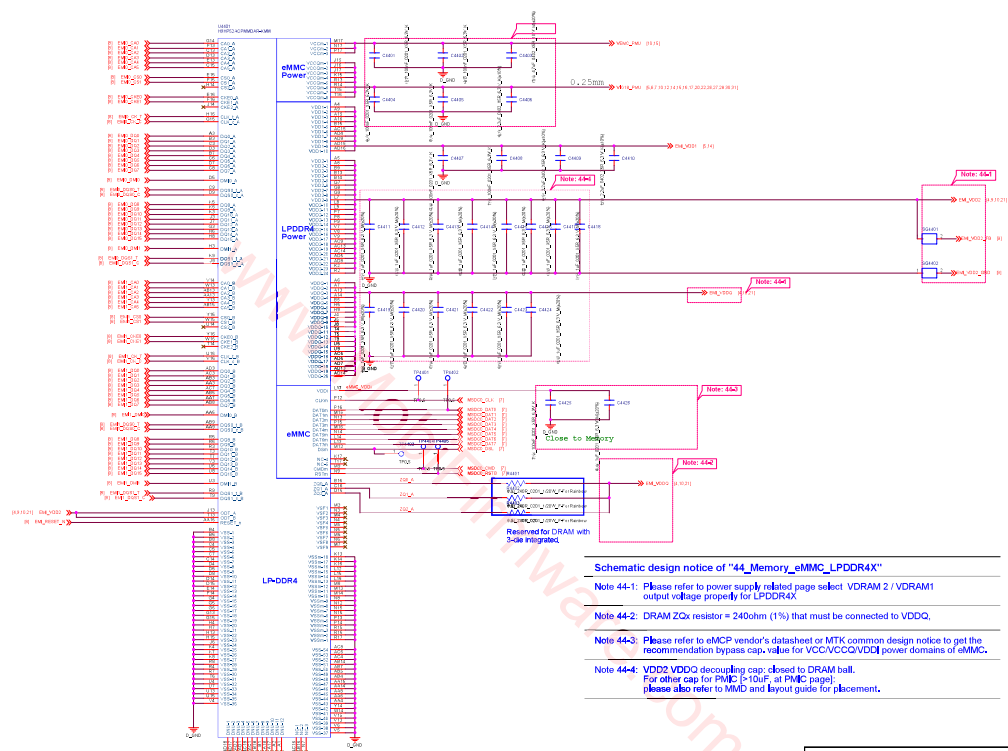


DIV_MB

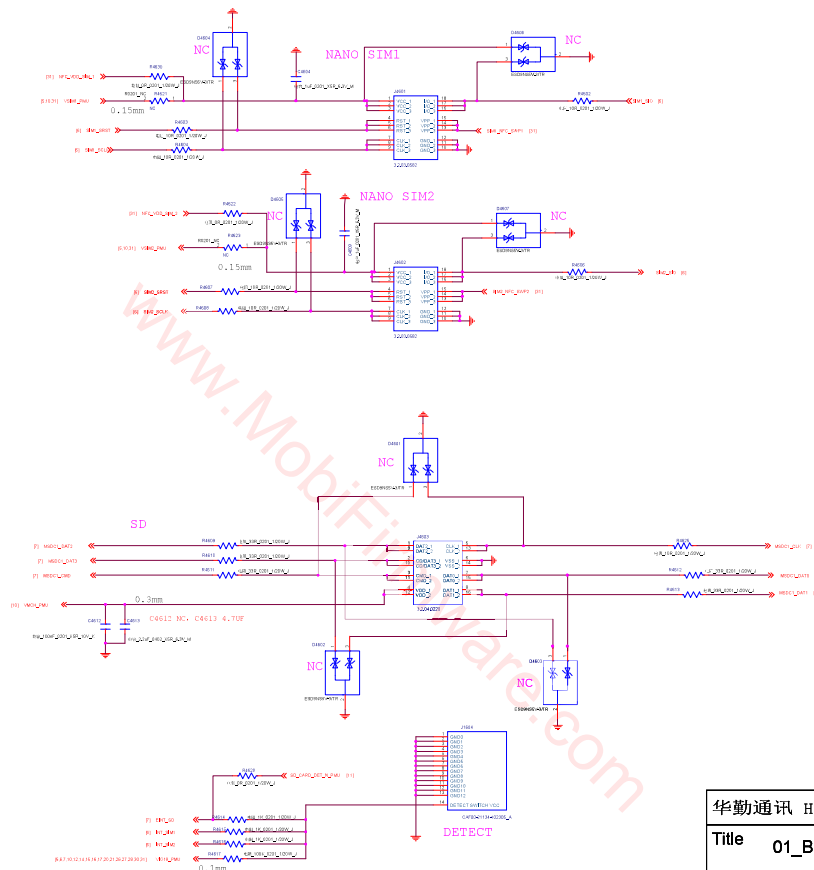


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Title 34_RF_MT6177_RF_DRX			
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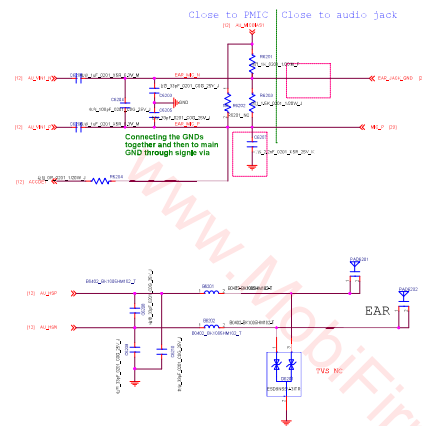
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Note 45-1:
Schematic design notice of "46_MEMORY_SD Card" page.
For better ESD performance, please select suitable components for system protection.

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Earphone Microphone



Schematic design notice of "62_PERI_AUDIO_IO" page.

Note 62-1: Part # of BEAD6202, BEAD6203, BEAD6204 and BEAD6205 needs changed to "BIM18BD102SN1" for high THD performance (-90dB) but this BOM change will results in FM RSSI 10dB degraded.

Note 62-2: Reserved Cap for CS/RS test, please double check multi-key function when used

Note 62-3:

Part #	Part #
BEAD6202	BIM18BD102SN1
BEAD6203	BIM18BD102SN1
BEAD6204	BIM18BD102SN1
BEAD6205	BIM18BD102SN1

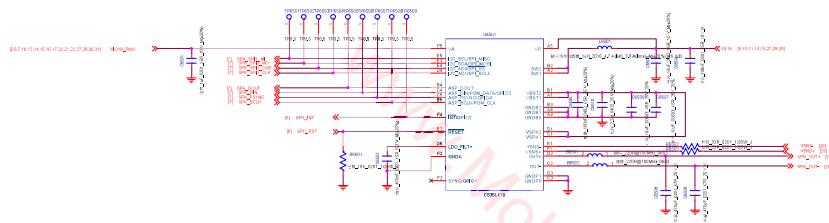
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Speaker Amplifier

MT6560 Speaker AMP I2C Address: 0x34 (Write:0x68, Read:0x69) when AD51 = GND; AD52 = GND.



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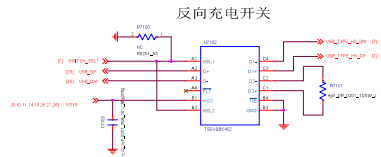
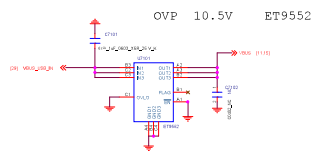
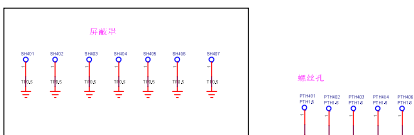
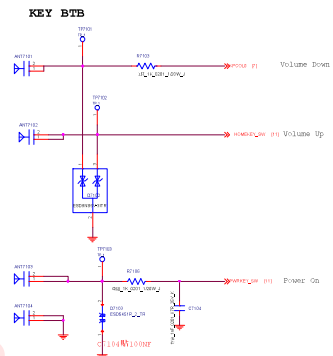
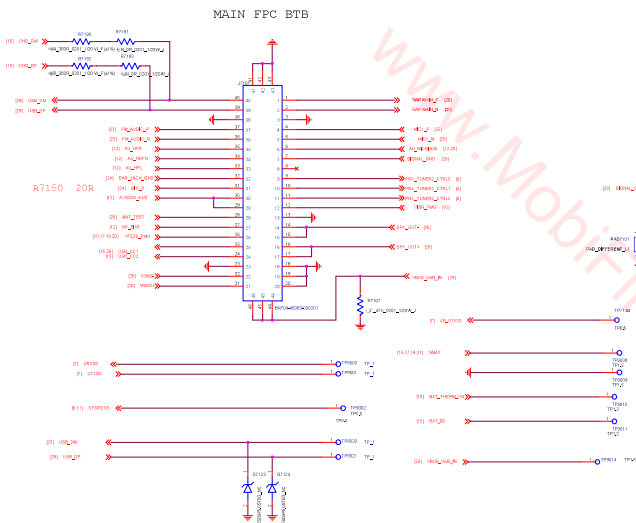


Table 1. Function Table

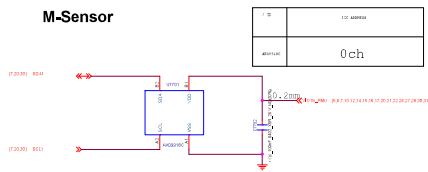
DE	SEL1	SEL2	D- Connection	D+ Connection
H	X	X	High-Z	High-Z
L	L	L	D- to D1-	D+ to D1+
L	L	H	D- to D1-	D+ to D2+
L	H	H	D- to D2-	D+ to D2+



USB

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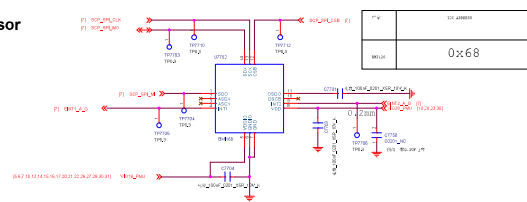
M-Sensor



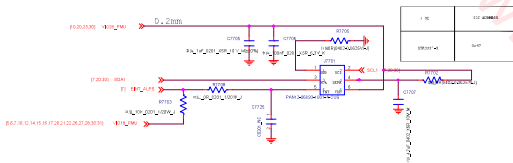
Accerometer + Gyro Sensor

default SPI (if sensor hub support)

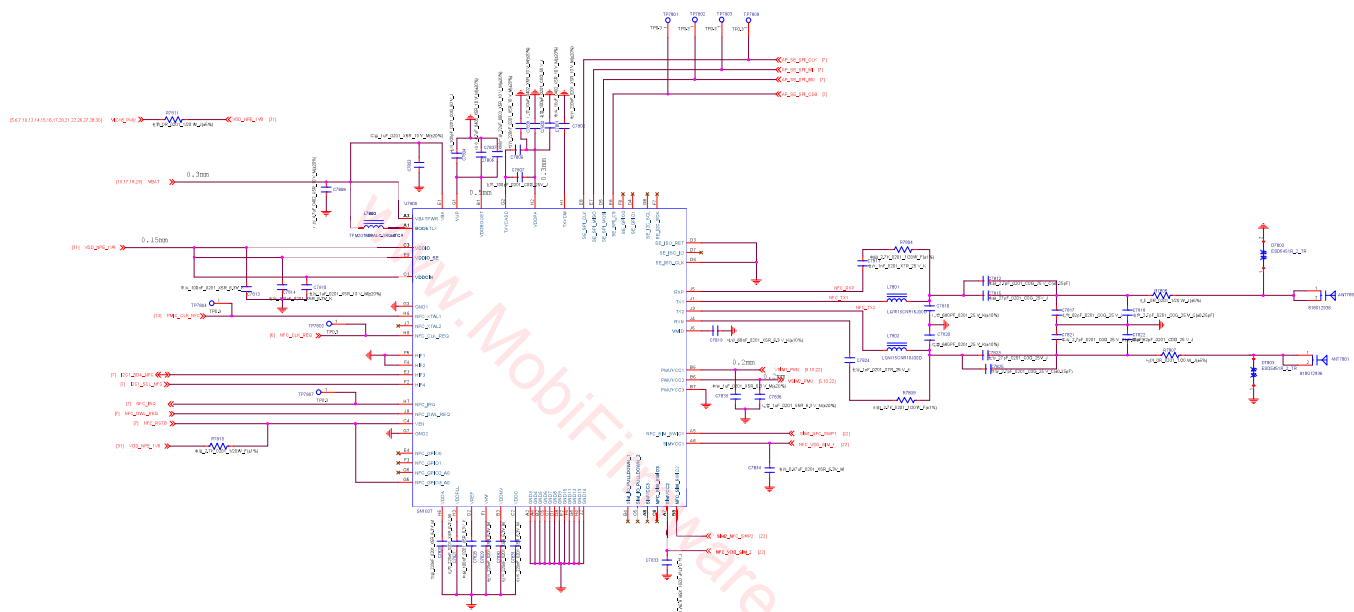
Gyro+A sensor



ALS PS STK3337-X



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